CSE274:APPLIED MACHINE LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain the types of data and data pre-processing concepts

CO2 :: apply feature engineering and dimensionality reduction techniques for machine learning models.

CO3 :: compare and Implement nonlinear and distance-based models using appropriate ROC- and precision-recall–based evaluation metrics.

CO4 :: apply linear, regularized, and tree-based regression models using standard regression performance metrics.

CO5 :: apply ensemble learning models through pipelines and systematic hyperparameter tuning techniques.

CO6 :: perform clustering using K-means, hierarchical, and density-based methods and evaluate results using established clustering metrics.

Unit I

Data Pre-processing : Types of data, Data pre-processing, data leakage concept, handling missing values, outliers handling,, handling categorical data, scaling and normalization, class imbalance handling

Unit II

Feature Engineering : variance threshold Feature Selection, correlation-based removal Feature Selection, forward selection, backward elimination, tree-based feature importance, Feature Extraction, creating new features, aggregation features

Dimensionality Reduction : Curse of dimensionality,, Principal Component Analysis, explained variance ration, Linear Discriminant Analysis

Unit III

Linear & Probabilistic Classification Models : Perceptron, Logistic Regression, Naïve Bayes Classifier, Model evaluation using confusion matrix, precision, recall, F1-Score, Introduction to supervised Learning

Nonlinear & Distance-Based Models : K-Nearest Neighbors, Decision Tree, Support Vector Machine, Model evaluation using ROC Curve, ROC-AUC, Precision-Recall Curve, AUC-PR

Unit IV

Regression : Difference between regression and classification, Bias-variance considerations in regression, Simple Linear Regression, Multiple Linear Regression, Interpretation of coefficients, Regularized Regression models,, Effect of regularization on model complexity, Polynomial feature expansion, Tree-Based regression models, Time-series Regression models

Unit V

Ensemble Learning : Introduction to ensemble learning, Bagging & Boosting Ensembles, majority voting classifier, Random Forest, AdaBoost, Gradient Boosting Machines, XGBoost, LightGBM, CatBoost, Ensemble Regression Models

Model evaluation and hyperparameter tuning : Pipelines, Cross-validation strategies, Grid Search, Random Search, Bayesian Optimization

Unit VI

Unsupervised Learning : Role of unsupervised learning, Differences between supervised and unsupervised learning, Euclidean, Manhattan, Cosine distances, Choosing appropriate distance metrics

Clustering Algorithms : K-Means Clustering, Elbow method, K-Medoids, Hierarchical Clustering, linkage criteria, Density-Based Clustering, Anomaly Detection, Evaluating clustering algorithms using Inertia, Silhouette Score, Davies–Bouldin Index

List of Practicals / Experiments:

List of Practicals

- Using a real-world dataset, perform data pre-processing by identifying data types, handling missing values, detecting and treating outliers, encoding categorical variables, and applying scaling and normalization techniques.

- Using a classification dataset, demonstrate handling of class imbalance using under-sampling, over-sampling, and SMOTE techniques, and illustrate data leakage scenarios with proper train–test splitting.
- Apply feature selection techniques including variance thresholding, correlation-based feature removal, forward selection, backward elimination, and tree-based feature importance on a given dataset.
- Implement dimensionality reduction using PCA and LDA on a high-dimensional dataset, analyze explained variance ratio, and visualize data in reduced dimensions.
- Implement Perceptron, Logistic Regression, and Naïve Bayes classifiers on a labeled dataset, and evaluate model performance using confusion matrix, precision, recall, and F1-score.
- Implement K-Nearest Neighbors, Decision Tree, and Support Vector Machine classifiers, and evaluate the models using ROC curve, ROC-AUC, Precision–Recall curve, and AUC-PR.
- Implement regularized regression , polynomial regression, and tree-based regression models, and evaluate performance using MAE, MSE, RMSE, R^2 score, residual plots, and actual vs predicted plots.
- Implement ensemble classification models including majority voting, Random Forest, AdaBoost, Gradient Boosting, XGBoost, LightGBM, and CatBoost, and compare their performance with individual base learners.
- Build machine learning pipelines and apply cross-validation, Grid Search, Random Search, and Bayesian Optimization to tune hyperparameters and improve model performance.
- Implement clustering algorithms including K-Means, K-Medoids, and Hierarchical Clustering, determine the optimal number of clusters using the Elbow method, and analyze linkage criteria.
- Apply DBSCAN for density-based clustering and anomaly detection, and evaluate clustering quality using Inertia, Silhouette Score, and Davies–Bouldin Index.

Text Books:

1. MACHINE LEARNING FOR ABSOLUTE BEGINNERS: A PLAIN ENGLISH INTRODUCTION (THIRD EDITION) by OLIVER THEOBALD, NULL

References:

1. MACHINE LEARNING by S SRIDHAR, M VIJAYALAKSHMI, OXFORD UNIVERSITY PRESS